Horowitz-Manski-Lee Bounds with Multilayered Sample Selection

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Motivation

- US government spends 19 billion dollars on 40 employment and training programs (Council of Economic Advisers 2019)
- Large literature estimates impact on employment and earnings
- Earnings effect reflect impact on wages (price effect) and employment (quantity effect)
- Different margins capture different mechanisms:
  - Wage effects = human capital
  - Labor supply effects = job search assistance
- Identifying causal effect of training on wages empirically challenging due to sample selection bias (Heckman 1979)
- Lee (2009): Partially identify wage effect

Motivation

- From a policy perspective, many job training programs are designed to get individuals a "good job"
  - e.g., Biden-Harris "Good Jobs Challenge"
- In standard model of labor market, firms do not come into play
- Accumulating evidence highlights role of firms in wage setting:
- "Firm effects" have been shown to matter for:
  - wage inequality, cyclicality of wages, early career progression, earnings losses of displaced workers, gender/racial wage gaps
- Does job training raise earnings by moving participants into higher-paying firms?
- Currently no framework to isolate the role of firms.
  - Lee (2009) derived bounds in a setting with no firm heterogeneity

This Paper

- Generalize the sample selection bias framework in Heckman (1979) and Lee (2009) by allowing for firm heterogeneity
- Show that causal effect of training on wages of "always employed" decomposes into:
  - **Within-firm effect**: contrast in wages between trained and untrained for fixed firm type
  - **Sorting effect**: contrast in wages between different firm types for fixed level of training
- Introduce two-step approach to derive analytic expressions for sharp bounds on causal effects of interest
- Bounds leverage empirical wage distributions for each firm in treatment and control groups
- Consider empirical application using Job Corps RCT and show that the bounds in Lee (2009) mostly capture sorting effect

Contribution to Literature

1. **Effectiveness of active labor market programs**
   - Heckman, Lalonde and Smith 1999; Schochet, Burghardt, and McConnell 2008; Lee 2009; Card, Kluve and Weber 2010, 2018; Katz et al 2022; Andersson et al forthcoming
2. **Firm-specific wage premia and labor market power**
   - AKM 1999; Card, Heining and Kline 2013; Song et al 2018; Bonhomme, Lamadon and Manresa 2019; Card, Cardoso, Heining and Kline 2018; Sorkin 2018; Lamadon, Mogstad and Setzler 2022; Berger, Herkenhoff and Mongey 2022; Kroft, Luo, Mogstad and Setzler 2023; Chan et al 2023
3. **Sample selection and treatment effects**
   - parametric and semi-parametric approaches: Heckman 1979, 1990; Ahn and Powell 1993; Andrews and Schafgans 1998; Das, Newey and Vella 2003
   - worst-case bounds: Horowitz and Manski 1995; Lee 2009; Semenova 2020
4. **Mediation analysis**
   - Pearl 2001; Bugni et al 2023; Roth and Kwon 2024

Outline of Talk

Review of Horowitz and Manski (1995) and Lee (2009)

Interpreting Lee (2009) Bounds in the Presence of Multilayered Sample Selection

Sharp Bounds on Direct Training Effects in the Multilayered Sample Selection Model

Simulations

Empirical application: Job Corps Program

Sample Selection Problem: Heckman (1979)

- Consider sample selection problem in Heckman (1979):

$$\begin{aligned} Y &= \begin{cases} \beta + \alpha Z + U, & \text{if } D = 1 \\ \text{unobserved}, & \text{if } D = 0 \end{cases} \\ D &= 1\{\theta + \eta Z + V > 0\} \end{aligned}$$

  where $(U, V) \perp Z$, $\alpha =$ wage effect, $\eta =$ labor supply effect

- Can classify different **response types**:
  1. *Always employed* **AE**: $\{V : V \geq -\theta\}$
  2. *Never employed* **NE**: $\{V : V \leq -\theta - \eta\}$
  3. *Compliers* **C**: $\{V : -\theta - \eta \leq V < -\theta\}$
  4. *Defiers* **Def**: $\{V : -\theta \leq V < -\theta - \eta\}$

- Monotonicity assumption: No Defiers, $\mathbb{P}(\mathbf{Def}) = 0$

Causal Effect of Training on Wages

- Causal effect of job training on wages given by α
- If we can identify the always-employed, $\{V : V \geq -\theta\}$:

$$\alpha = \mathbb{E}[Y|Z=1, V \geq -\theta] - \mathbb{E}[Y|Z=0, V \geq -\theta]$$

- Monotonicity $\rightarrow \mathbb{E}[Y|Z=0, D=1] = \mathbb{E}[Y|Z=0, V \geq -\theta]$
- However, $\mathbb{E}[Y|Z=1, D=1] \neq \mathbb{E}[Y|Z=1, V \geq -\theta]$
  - $\{Z=1, D=1\}$ consists of always-employed **and compliers**
  - Sample selection problem: presence of compliers
- Lee (2009): partially identify $\mathbb{E}[Y|Z=1, V \geq -\theta]$ using Horowitz and Manski (1995) "trimming" bounds

Horowitz and Manski (1995) “Trimming” Bounds

- Consider mixture distribution $F(y) = \gamma F_{Y_1} + (1-\gamma)F_{Y_2}$
- When only $Y \sim F$ is observed, Horowitz and Manski (1995) Bounds on $E[Y_1]$ for known γ:

$$\mathbb{E}[Y|Y \leq F^{-1}(\gamma)] \leq E[Y_1] \leq \mathbb{E}[Y|Y \geq F^{-1}(1-\gamma)]$$

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- **Lee (2009)**: apply this idea to the mixture

$$F_{Y|Z=1,D=1} = \gamma F_{Y|Z=1,\mathbf{AE}} + (1-\gamma)F_{Y|Z=1,\mathbf{C}}$$

- Under RA+Mon, $\gamma \equiv \frac{\mathbb{P}[\mathbf{AE}]}{\mathbb{P}[\mathbf{AE} \cup \mathbf{C}]} = \frac{\mathbb{P}[D=1|Z=0]}{\mathbb{P}[D=1|Z=1]}$ is point identified
- The resulting bounds on α are

$$\mathbb{E}[Y|Z=1, D=1, Y \leq F^{-1}_{Y|Z=1,D=1}(\gamma)] - \mathbb{E}[Y|Z=0, D=1]$$
$$\leq \alpha \leq$$
$$\mathbb{E}[Y|Z=1, D=1, Y \geq F^{-1}_{Y|Z=1,D=1}(1-\gamma)] - \mathbb{E}[Y|Z=0, D=1]$$

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Multilayered Sample Selection: Potential Outcomes

- $Y_{z,d}(\omega)$ is potential outcome if individual ω was assigned to treatment group $z \in \{0, 1\}$ and is employed at firm $d \in \{1, ..., K\}$
- $D_z(\omega) \in \{0, 1, \ldots, K\}$ is the potential employer (with 0 denoting unemployment) if individual ω is assigned $Z = z$
- Z is observed treatment status and D observed employer
- Relationship between observed and potential outcomes:

$$Y = \sum_{d=1}^{K} [Y_{1,d} Z + Y_{0,d}(1 - Z)] 1\{D = d\}$$

$$D = D_1 Z + D_0(1 - Z)$$

DAG Representation: Lee's model

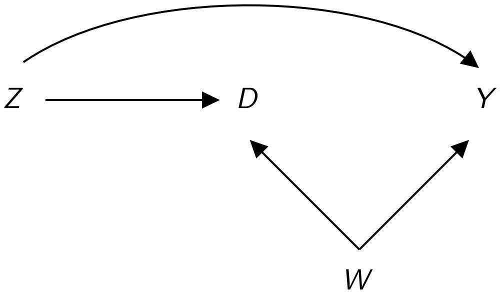

Figure: DAG when there are no firm effects.

- Assumption: $Y_{z,d} = Y_z$
- $\mathbf{AE} = \{D_0 > 0, D_1 > 1\}$
- Parameter of interest: $\mathbb{E}[Y_{z=1} - Y_{z=0} | \mathbf{AE}]$

DAG Representation

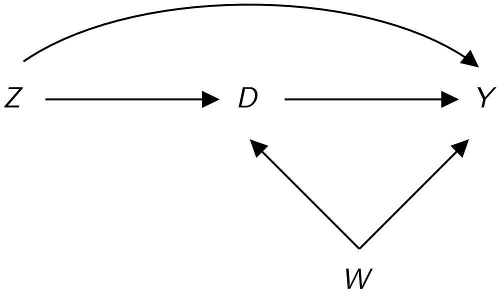

Figure: our model.

- $Y_{z,d} \neq Y_z$
- Define *response type* $T = (D_0, D_1)$.
- Parameter of interest: $\mathbb{E}[Y_{1,d} - Y_{0,d} | T = (d, d)]$

Two Key Assumptions

- **Assumption 1** *Random Assignment*

$$\{(Y_{z,d}, D_z) : d \in \{1, ..., K\}, z \in \{0, 1\}\} \perp Z$$

- Individuals are randomly assigned to a treatment group
- **Assumption 2** *Lee's Monotonicity*

$$\mathbb{P}(\mathbb{1}\{D_1 > 0\} \geq \mathbb{1}\{D_0 > 0\}) = 1$$

- Being assigned to treatment group never lowers employment

Decomposing Wage Effect for Always Employed

"Always Employed" (**AE**): those with $\{D_0 > 0, D_1 > 0\}$.

- When $Y_{z,d} \neq Y_z$, the wage effect for **AE** can be decomposed:

$$
\begin{aligned}
&\mathbb{E}[Y_{1,D_1} - Y_{0,D_0}|\mathbf{AE}] \\
&= \underbrace{\sum_{d=1,d'=1}^{K,K} \mathbb{E}[Y_{1,d} - Y_{0,d}|T = (d', d)]\mathbb{P}(T = (d', d)|\mathbf{AE})}_{\mathbb{E}[Y_{1,D_1} - Y_{0,D_1}|\mathbf{AE}]} \\
&\quad + \underbrace{\sum_{\substack{d=1,d'=1: \\ d \neq d'}}^{K,K} \mathbb{E}[Y_{0,d} - Y_{0,d'}|T = (d', d)]\mathbb{P}(T = (d', d)|\mathbf{AE})}_{\mathbb{E}[Y_{0,D_1} - Y_{0,D_0}|\mathbf{AE}]}
\end{aligned}
$$

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- Interpreting Lee (2009) Bounds in the Presence of Multilayered Sample Selection
- Sharp Bounds on Direct Training Effects in the Multilayered Sample Selection Model
- Simulations
- Empirical application: Job Corps Program

Causal Effect of Job Training

- Target parameter is $\mathbb{E}[Y_{1,d} - Y_{0,d} | T = (d, d)]$.
- Consider two firm types so that $D \in \mathcal{D} \equiv \{0, L, H\}$
- Under Assumption 2, the set of response types are:

$$\{(0,0),(0,L),(0,H),\underbrace{(L,L),(H,H),(L,H),(H,L)}_{\textbf{AE}}\}$$

- Contrast with Lee (2009): $\{(0,0),(0,1),\underbrace{(1,1)}_{\textbf{AE}}\}$

General Identification Approach

Under Assumption 1, we have that the observed densities, $f_{Y,D=d|Z=z}$, satisfy the following system of equations

$$
\begin{aligned}
f_{Y,D=d|Z=1}(y) &= \sum_{d'=0}^{K} \mathbb{P}[T=(d',d)] \times f_{Y_{1,d}|T}(y|d',d) \quad \forall d,y \\
f_{Y,D=d|Z=0}(y) &= \sum_{d'=0}^{K} \mathbb{P}[T=(d,d')] \times f_{Y_{0,d}|T}(y|d,d') \quad \forall d,y
\end{aligned}
$$

- In fact, the empirical content of the multilayered sample selection model is characterized by these equations.

General Identification Approach

Step 1: Identified set for Response type probabilities is exactly the set of non-negative solutions to the following linear system

$$\mathbb{P}(D = d|Z = 1) = \sum_{d'=0}^{K} \mathbb{P}[T = (d', d)] \quad \forall d \tag{1}$$

$$\mathbb{P}(D = d|Z = 0) = \sum_{d'=0}^{K} \mathbb{P}[T = (d, d')] \quad \forall d \tag{2}$$

- $\Theta_I \equiv \{(\mathbb{P}[T = (d, d')] : d, d' \in \mathcal{D}) \mid \text{(1-2) holds}\}$.
- Obtained by integrating y out of the earlier system.
  - The lack of an exclusion restriction allows us to do this without losing information about the response-type probabilities
  - In general, the identified set for response-type probabilities can depend on the outcome distribution; see Vayalinkal (2024)
- Add'l restrictions on these probabilities can be added easily.

General Identification Approach

Define

$$\underline{\gamma}_d \equiv \inf_{\Theta_I}\{\mathbb{P}[T = (d,d)]\}$$

Step 2: Sharp bounds on $\mathbb{E}[Y_{1,d} - Y_{0,d} | T = (d,d)]$.

- Lower bound

$$\mathbb{E}\left[Y|D = d, Z = 1, Y \leq F^{-1}_{Y|D=d,Z=1}\left(\frac{\underline{\gamma}_d}{\mathbb{P}(D = d|Z = 1)}\right)\right] - \mathbb{E}\left[Y|D = d, Z = 0, Y \geq F^{-1}_{Y|D=d,Z=0}\left(1 - \frac{\underline{\gamma}_d}{\mathbb{P}(D = d|Z = 0)}\right)\right]$$

- Upper bound

$$\mathbb{E}\left[Y|D = d, Z = 1, Y \geq F^{-1}_{Y|D=d,Z=1}\left(1 - \frac{\underline{\gamma}_d}{\mathbb{P}(D = d|Z = 1)}\right)\right] - \mathbb{E}\left[Y|D = d, Z = 0, Y \leq F^{-1}_{Y|D=d,Z=0}\left(\frac{\underline{\gamma}_d}{\mathbb{P}(D = d|Z = 0)}\right)\right]$$

General Identification Approach

Lemma

Suppose Assumptions 1 and 2 hold, and assume that there are only 2 firm types. Then, for any $d, d' \in \{H, L\}$ with $d \neq d'$, we have

$$\underline{\gamma}_d = max\{0, \mathbb{P}(D = d|Z = 0) - \mathbb{P}(D = d'|Z = 1)\}$$

- Can impose restrictions on $\mathbb{P}(T = (H, L))$ to tighten bounds by making $\underline{\gamma}_d$ larger:
  1. $\mathbb{P}(T = (H, H)) \geq \mathbb{P}(T = (H, L))$
  2. $\mathbb{P}(T = (L, L)) \geq \mathbb{P}(T = (H, L))$
  3. $\min_\tau \mathbb{P}(T = \tau) = \mathbb{P}(T = (H, L))$
  4. $\mathbb{P}(T = (H, L)) = 0$

  Expressions for $\underline{\gamma}_d$

- Sharp falsification test: model + assumptions jointly rejected by data iff identified set for response-type probabilities is empty. Doesn't depend on Y distribution.

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Simulations

- Fix the following response-type probabilities in identified set:

$$\mathbb{P}(T = (H, H)) = 0.37$$
$$\mathbb{P}(T = (L, L)) = 0.28$$

- This fixes all other response-type probabilities
  - $\mathbb{P}(T = (0, 0)) = 0.29$
  - $\mathbb{P}(T = (0, L)) = 0.02$
  - $\mathbb{P}(T = (0, H)) = 0.00$
  - $\mathbb{P}(T = (L, H)) = 0.04$
  - $\mathbb{P}(T = (H, L)) = 0.00$

System of Equations

Simulation 1: No Direct (Same-Firm) Effect of Training

The conditional distributions used in this simulation are:

t	Dist. of $\exp(Y_{1,D_1})\|T = t$	Dist. of $\exp(Y_{0,D_0})\|T = t)$
$(0, L)$	Lognormal$(9.5, 1)$	Lognormal$(9.5, 1)$
$(0, H)$	Lognormal$(11.5, 1)$	Lognormal$(9.5, 1)$
(L, H)	Lognormal$(16.5, 1)$	Lognormal$(9.5, 1)$
(H, L)	Lognormal$(9.75, 1)$	Lognormal$(9.6, 1)$
(L, L)	Lognormal$(9.5, 1)$	Lognormal$(9.5, 1)$
(H, H)	Lognormal$(14.5, 1)$	Lognormal$(14.5, 1)$

Table: DGP for Simulation 1

Simulation 2: Positive Direct (Same-Firm) Effect

The conditional distributions used in this simulation are:

t	Dist. of $\exp(Y_{1,D_1})\|T=t$	Dist. of $\exp(Y_{0,D_0})\|T=t)$
$(0, L)$	Lognormal$(10.5, 1)$	Lognormal$(9.5, 1)$
$(0, H)$	Lognormal$(12.5, 1)$	Lognormal$(9.5, 1)$
(L, H)	Lognormal$(14.5, 1)$	Lognormal$(9.5, 1)$
(H, L)	Lognormal$(10.5, 1)$	Lognormal$(10.5, 1)$
(L, L)	Lognormal$(10.5, 1)$	Lognormal$(9.5, 1)$
(H, H)	Lognormal$(14, 1)$	Lognormal$(12, 1)$

Table: DGP for Simulation 2

Simulation (1 & 2): Identified Set for Type Probabilities

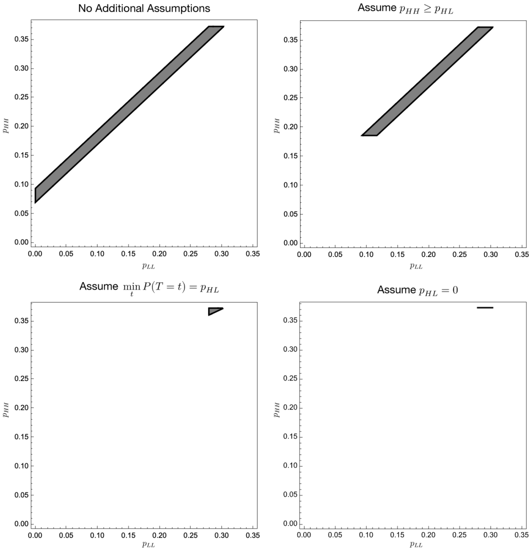

Figure: Identified Set for $\Big(\mathbb{P}\big(T = (L, L)\big), \mathbb{P}\big(T = (H, H)\big)\Big)$

Simulation 1: No Direct (Same-Firm) Effect of Training

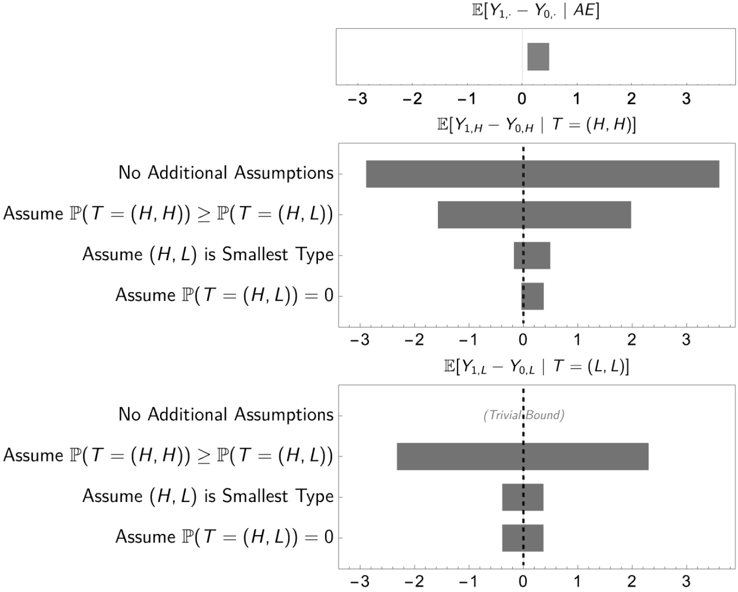

Figure: Dashed line is True DGP's value of parameter

Simulation 2: Positive Direct Effect of Training

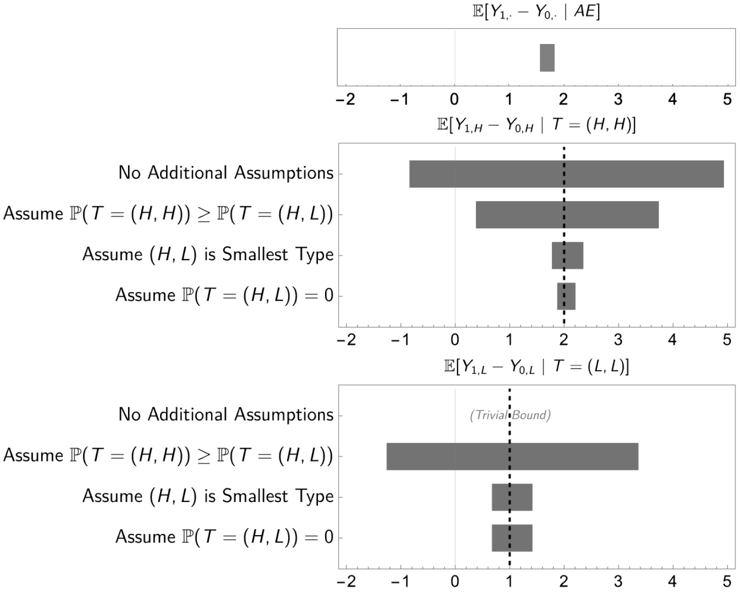

Figure: Dashed line is True DGP's value of parameter

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Job Corps Program

1. **Background**
   - Largest residential career training program in the U.S.
     - 110 Job Corps centers at time of study (131 Job Corps centers today)
     - Has trained $>$ 2 million individuals since inception in 1964 (60,000 trainees/year today)
     - Provision cost of 14,000 USD/enrollee at time of study (34,000 USD/enrollee today)
   - Key features:
     - Targets disadvantaged 16-24 year olds
     - Free for trainees
     - Trainees live at local Job Corps center during program
2. **Curriculum**
   - 30 weeks in length and contains two core components:
     - Training: 440 hours academic instruction, 700 hours vocational training
     - Job search assistance at exit

Job Corps Study

1. **Overview**
   - U.S. Department of Labor funded
   - Eligibility:
     - First-time applicants between November 1994 and December 1995: 80,883 individuals
   - Control group embargoed from participating in Job Corps
2. **Research Sample**
   - Eligible applicants randomly assigned:
     - $N_c = 5{,}977$
     - $N_t = 9{,}409$
     - Remainder to program non-research group
   - $N = 15{,}386$

Sample Restrictions and Key Variables

1. **Sample Restrictions**
   - Lee (2009) sample: $N = 9{,}145$
     - Excludes if missing weekly earnings/hours in any of 208 weeks[1]
   - Sample with non-missing health amenity: $N = 6{,}403$
     - Excludes if employed and missing health insurance amenity observation at weeks of interest: 90, 135, 180 and 208
2. **Key Variables**
   - Employment
   - Hourly wage (= weekly earnings / weekly hours worked)
   - Amenities: Health insurance, paid sick leave, paid vacation, child care assistance, flexible hours, employer-provided transportation, retirement or pension benefits, dental plan, tuition reimbursement or training course

[1] Follows Lee (2009).

Summary Statistics by Treatment Status

	Control		Treated		Difference	
	Mean	S.D.	Mean	S.D.	Difference	S.E.
White, non-Hispanic	0.26	0.44	0.27	0.44	0.00	0.01
Black, non-Hispanic	0.49	0.50	0.49	0.50	0.00	0.01
Hispanic	0.17	0.38	0.17	0.37	-0.00	0.01
Other race/ethnicity	0.07	0.26	0.07	0.26	-0.00	0.01
Education	10.11	1.54	10.11	1.56	0.01	0.03
At baseline						
Have job	0.19	0.39	0.20	0.40	0.01	0.01
Mths. empl. prev. yr.	3.53	4.24	3.60	4.25	0.07	0.09
Had job, prev. yr.	0.63	0.48	0.63	0.48	0.01	0.01
Earnings, prev. yr.	2810.48	4435.62	2906.45	6401.33	95.97	117.10
Usual hours/week	20.91	20.70	21.82	21.05	0.91	0.45
Usual weekly earn.	102.89	116.46	110.99	350.61	8.10	5.09
Post randomization						
Week 52 hours	17.78	23.39	15.30	22.68	-2.49	0.49
Week 104 hours	21.98	26.08	22.64	26.25	0.67	0.56
Week 156 hours	23.88	26.15	25.88	26.57	2.00	0.56
Week 208 hours	25.83	26.25	27.79	25.74	1.95	0.56
Week 52 earn.	103.80	159.89	91.55	149.28	-12.25	3.33
Week 104 earn.	150.41	210.24	157.42	200.27	7.02	4.42
Week 156 earn.	180.88	224.43	203.71	239.80	22.84	4.94
Week 208 earn.	200.50	230.66	227.91	250.22	27.41	5.11
Total 4 yr. earn.	30006.69	26893.60	30800.41	26437.39	793.72	571.83
Sample size	3599		5546		9145	

Table: Summary statistics by treatment status[2]

[2] Weekly earnings not conditional on employment.

Amenities by Treatment Status Among Employed

	Control		Treated		Difference	
	Mean	S.D.	Mean	S.D.	Difference	S.E.
Health insurance	0.49	0.50	0.51	0.50	0.02	0.02
Paid sick leave	0.39	0.49	0.43	0.49	0.03	0.02
Paid vacation	0.54	0.50	0.58	0.49	0.04	0.02
Childcare assistance	0.13	0.34	0.14	0.35	0.01	0.01
Flexible hours	0.53	0.50	0.56	0.50	0.02	0.02
Employer-provided transportation	0.20	0.40	0.19	0.39	-0.01	0.01
Pension or retirement benefits	0.37	0.48	0.39	0.49	0.03	0.02
Dental plan	0.39	0.49	0.43	0.50	0.04	0.02
Tuition reimbursement	0.22	0.42	0.26	0.44	0.04	0.02
Employment	0.44	0.50	0.44	0.50	0.00	0.01
Share Employment Full-Time ($>$ 30h/week)	0.83	0.37	0.85	0.36	0.02	0.01
Sample size	3288		5091		8379	

Table: Probability of working at amenity-providing firm (week 90)[3]

Probability of working at amenity-providing firm (week 208) Distribution of amenity types (week 90)

[3] Conditional on employment.

Mean Wages Across Firm-Types

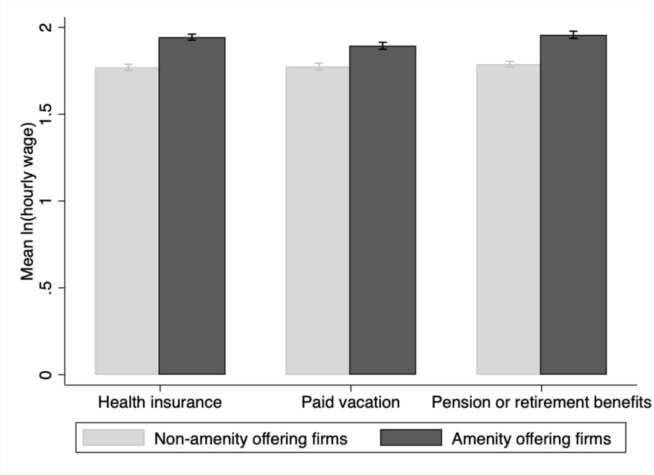

Figure: Mean ln(hourly wage) by firm amenity provision (week 90)

Mean ln(hourly wage) by firm amenity provision (week 208)

CDF ln(hourly) wage by firm type and treatment status (week 90)

Sharp bounds

- Assume that firm type d is observable:
  - $d = H$ if firm offers health insurance
  - $d = L$ if firm does not offer health insurance
- First report bounds for $\mathbb{P}(L, L)$ and $\mathbb{P}(H, H)$ in base case
  - and under several restrictions on $\mathbb{P}(H, L)$
- Next report multilayered bounds
- Focus on results at week 90

Identified Set for Type Probabilities (Week 90)

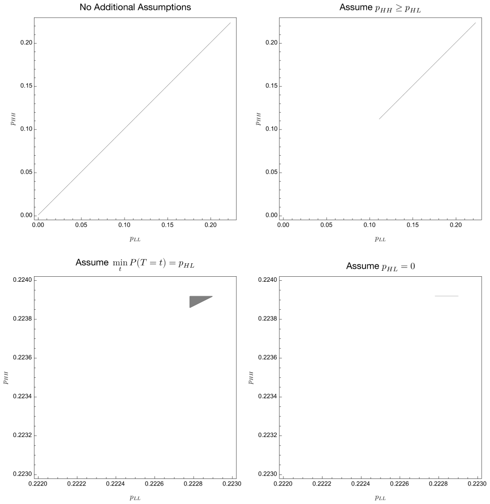

Figure: Identified set for $\left(\mathbb{P}(T = (L, L)), \mathbb{P}(T = (H, H))\right)$ (week 90)

Identified set (week 208) Propensity scores by week

Job Corps (Week 90)

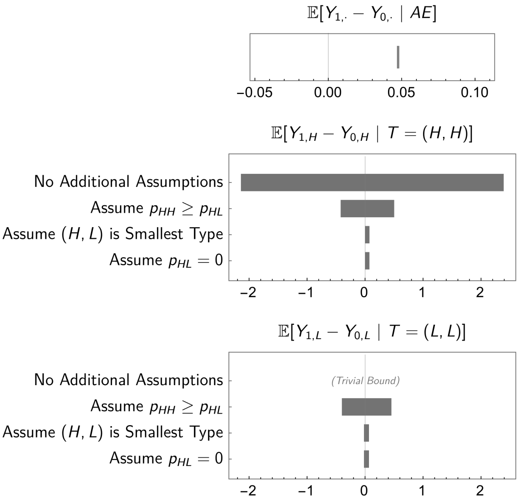

Figure: Bounds for $\mathbb{E}[Y_{1,d} - Y_{0,d} | T = (d, d)]$ (week 90)

Bounds (week 208) Replication of Lee (2009) Bounds

Conclusion

- Develop framework to generalize Heckman sample selection problem allowing for firm heterogeneity
- Show that "Lee bounds" partially identify mix of direct and indirect effects of training
- Derive closed-form, sharp bounds on direct effects
- Consider simulations and an empirical application to illustrate how bounds can be implemented empirically
- Future work:
  - NEW job training RCT in Canada with linkage to administrative employee-employer data and survey data on amenities
  - NEW structural job search model with amenities and production complementarities – decompose treatment into search and human capital effects
  - NEW MDRC WorkAdvance sector employment program RCT data including survey data on industry and occupation with linkage to administrative income data

Thank you!

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'Top 5' Papers featuring Multilayered Selection

	Number of papers	
Lee (2009) citations	Estimating bounds	Simplifying sample selection
56	42	7

Examples of papers simplifying sample selection

Bianchi and Giorcelli (2022), Giorcelli (2019)
- Treatment: Management training
- Outcome: Firm total factor productivity
- Selection addressed: Treatment affects firm survival
- Additional layer: Treatment affects industry selection

Cullen and Pakzad-Hurson (2023)
- Treatment: Private sector pay transparency laws
- Outcome: Worker wages
- Selection addressed: Treatment affects sorting into private sector
- Additional layer: Treatment affects sorting across private sector firms

Table: 'Top 5' papers simplifying sample selection to a single dimension to apply Lee bounds[4]

[4] 'Top 5' journals include: the American Economic Review, Econometrica, the Journal of Political Economy, the Quarterly Journal of Economics and the Review of Economic Studies.

Illustrating Horowitz-Manski Bounds

Suppose $F_Y(y) = \gamma F_{Y_1} + (1-\gamma) F_{Y_2}$ is a mixture distribution with unobserved component distributions F_{Y_1} and F_{Y_2} and weight $\gamma = 0.1$

Suppose quantile function of mixture distribution F_Y is as follows:

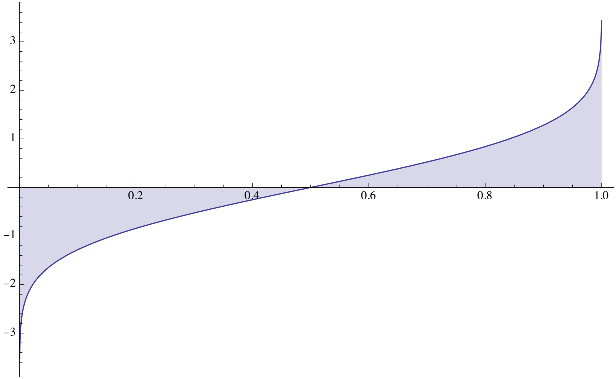
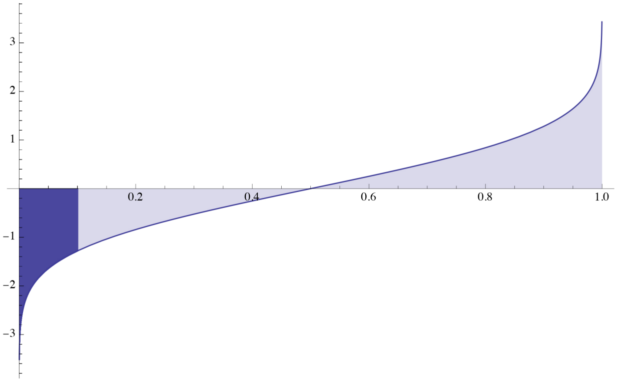


Figure: Quantile of F

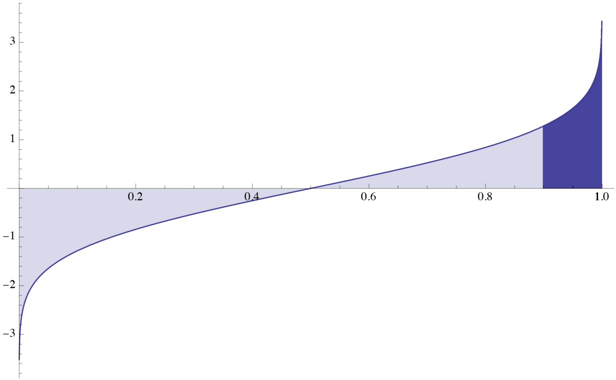
Illustrating the Horowitz-Manski Bounds

The lower bound for the mean of F_{Y_1} is given by integrating the left tail up to mass 0.1 and normalizing it to have total mass 1 (i.e. the mean outcome of the worst 10%):



Illustrating the Horowitz-Manski Bounds

The upper bound for the mean of F_{Y_1} is given by integrating the right tail up to mass 0.1 (i.e the best 10%):



Horowitz and Manski Bounds

System of Equations for Response-Type Probabilities

Solution can be obtained from the following system of equations:

$$\mathbb{P}\left(T=(0,0)\right)=1-\mathbb{P}(D=H|Z=1)-\mathbb{P}(D=L|Z=1)$$
$$\mathbb{P}\left(T=(0,L)\right)=\mathbb{P}\left(D=L|Z=1\right)-\left(\mathbb{P}\left(D=H|Z=0\right)+\mathbb{P}(T=(H,H))\right)-\mathbb{P}(T=(L,L))$$
$$\mathbb{P}\left(T=(0,H)\right)=\mathbb{P}\left(D=H|Z=1\right)-\left(\mathbb{P}\left(D=L|Z=0\right)+\mathbb{P}(T=(L,L))\right)-\mathbb{P}(T=(H,H))$$
$$\mathbb{P}\left(T=(L,H)\right)=\mathbb{P}(D=L|Z=0)-\mathbb{P}(T=(L,L))$$
$$\mathbb{P}\left(T=(H,L)\right)=\mathbb{P}(D=H|Z=0)-\mathbb{P}(T=(H,H))$$
$$\max\{0,\mathbb{P}(D=H|Z=0)-\mathbb{P}(D=L|Z=1)\}\leq\mathbb{P}(T=(H,H))\leq$$
$$\min\{\mathbb{P}(D=H|Z=0),\mathbb{P}(D=H,Z=1)\}$$
$$\max\{0,\mathbb{P}(D=L|Z=0)-\mathbb{P}(D=H|Z=1)\}\leq P(T=(L,L))\leq$$
$$\min\{\mathbb{P}(D=L|Z=0),\mathbb{P}(D=L,Z=1)\}$$
$$\mathbb{P}(D=0|Z=1)\leq\mathbb{P}(D=0|Z=0)$$
$$\mathbb{P}(D\neq 0|Z=0)\leq\mathbb{P}(D\neq 0|Z=1)$$

◂ Simulation Response-Type Probabilities

Expressions for $\underline{\gamma}_d$ Under Various Restrictions

- Under $\mathbb{P}(T=(H,H)) \geq \mathbb{P}(T=(H,L))$:

$$\underline{\gamma}_H = \max\left\{\frac{\mathbb{P}(D=H \mid Z=0)}{2}, \mathbb{P}(D=H \mid Z=0) - \mathbb{P}(D=L \mid Z=1)\right\}$$

$$\underline{\gamma}_L = \max\left\{0, \frac{\mathbb{P}(D=H \mid Z=0)}{2} - \mathbb{P}(D=H \mid Z=1) + \mathbb{P}(D=L \mid Z=0)\right\}$$

- Under $\mathbb{P}(T=(L,L)) \geq \mathbb{P}(T=(H,L))$:

$$\underline{\gamma}_H = \max\left\{0, \Big(\mathbb{P}(D=H \mid Z=0) - \mathbb{P}(D=L \mid Z=0)\Big), \left(\mathbb{P}(D=H \mid Z=0) - \frac{\mathbb{P}(D=L \mid Z=1)}{2}\right)\right\}$$

$$\underline{\gamma}_L = \max\left\{0, \left(\frac{\mathbb{P}(D=H \mid Z=0) + \mathbb{P}(D=L \mid Z=0) - \mathbb{P}(D=H \mid Z=1)}{2}\right), \Big(\mathbb{P}(D=L \mid Z=0) - \mathbb{P}(D=H \mid Z=1)\Big)\right\}$$

Expressions for $\underline{\gamma}_d$ Under Various Restrictions

- Under $\min_\tau \mathbb{P}(T=\tau) = \mathbb{P}(T=(H,L))$:

$$\underline{\gamma}_H = \max\left\{ \frac{\mathbb{P}(D=H \mid Z=0)}{2}, \right.$$
$$2\mathbb{P}(D=H \mid Z=0) - \mathbb{P}(D=H \mid Z=1),$$
$$\mathbb{P}(D=H \mid Z=0) - \frac{\mathbb{P}(D=L \mid Z=0)}{2},$$
$$\frac{3\mathbb{P}(D=H \mid Z=0) + \mathbb{P}(D=L \mid Z=0) - \mathbb{P}(D=H \mid Z=1) - \mathbb{P}(D=L \mid Z=1)}{2},$$
$$\mathbb{P}(D=H \mid Z=0) - \frac{\mathbb{P}(D=L \mid Z=1)}{3},$$
$$\left. \mathbb{P}(D=H \mid Z=0) + \mathbb{P}(D=H \mid Z=1) + \mathbb{P}(D=L \mid Z=1) - 1 \right\}$$
$$\underline{\gamma}_L = \max\left\{ 0, \mathbb{P}(D=H \mid Z=0) - \mathbb{P}(D=H \mid Z=1) + \mathbb{P}(D=L \mid Z=0) \right\}$$

- Under $\mathbb{P}(T=(H,L)) = 0$:

$$\underline{\gamma}_H = \mathbb{P}(D=H \mid Z=0)$$
$$\underline{\gamma}_L = \max\{0, \mathbb{P}(D=H \mid Z=0) - \mathbb{P}(D=H \mid Z=1) + \mathbb{P}(D=L \mid Z=0)\}$$

◂ Restrictions on $\mathbb{P}(T=(H,L))$

Amenities by Treatment Status Among Employed

	Control		Treated		Difference	
	Mean	S.D.	Mean	S.D.	Difference	S.E.
Health insurance	0.56	0.50	0.59	0.49	0.03	0.02
Paid sick leave	0.48	0.50	0.52	0.50	0.04	0.02
Paid vacation	0.64	0.48	0.65	0.48	0.01	0.01
Childcare assistance	0.14	0.35	0.17	0.37	0.02	0.01
Flexible hours	0.60	0.49	0.61	0.49	0.00	0.01
Employer-provided transportation	0.19	0.39	0.20	0.40	0.02	0.01
Pension or retirement benefits	0.45	0.50	0.49	0.50	0.04	0.02
Dental plan	0.49	0.50	0.52	0.50	0.03	0.02
Tuition reimbursement	0.30	0.46	0.31	0.46	0.02	0.01
Employment	0.55	0.50	0.59	0.49	0.04	0.01
Share Employment Full-Time ($>$ 30h/week)	0.85	0.36	0.88	0.33	0.03	0.01
Sample size	3288		5091		8379	

Table: Probability of working at amenity-providing firm (week 208)[5]

◂ Probability of working at amenity-providing firm (week 90)

[5] Conditional on employment.

Distribution of Types Among Employed

	Control		Treated		Difference		Pooled	
	Mean	S.D.	Mean	S.D.	Difference	S.E.	ln(Hourly wage)	S.D.
H=1, R=1, V=1	0.29	0.45	0.34	0.47	0.04	0.02	1.98	0.32
H=1, R=1, V=0	0.03	0.17	0.02	0.14	-0.01	0.01	1.93	0.31
H=1, R=0, V=1	0.11	0.32	0.11	0.31	-0.01	0.01	1.90	0.25
H=1, R=0, V=0	0.05	0.22	0.05	0.21	-0.00	0.01	1.82	0.28
H=0, R=1, V=1	0.03	0.17	0.03	0.18	0.01	0.01	1.88	0.30
H=0, R=1, V=0	0.02	0.14	0.01	0.11	-0.01	0.01	1.81	0.22
H=0, R=0, V=1	0.08	0.26	0.09	0.29	0.02	0.01	1.76	0.31
H=0, R=0, V=0	0.39	0.49	0.35	0.48	-0.04	0.02	1.76	0.31
Sample size	2454		3778				6232	

Table: Distribution of job types (week 90)[6]

◂ Probability of working at amenity-providing firm (Week 90)

[6] Conditional on employment.

Mean Wages Across Firm-Types

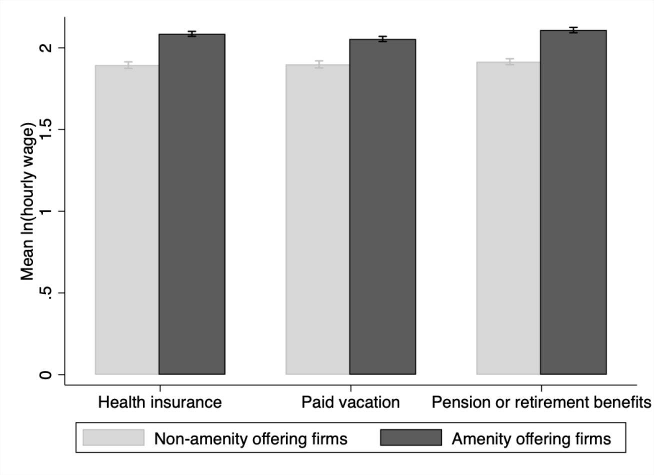
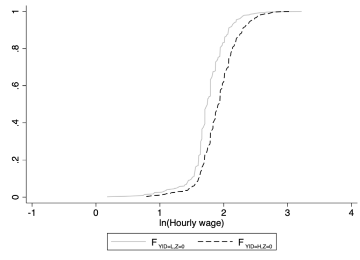


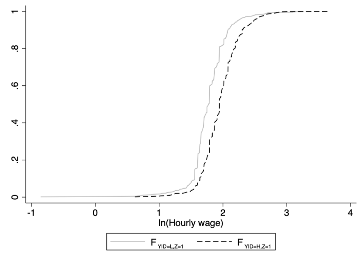
Figure: Mean ln(hourly wage) by firm amenity provision (week 208)

◂ Mean ln(hourly wage) by firm amenity provision (week 90)

Distribution of Wages Across Firm-Types by Treatment Status



(a) control units



(b) treated units

Figure: Empirical CDF of ln(hourly wage) by treatment status (week 90, amenity=health)

◂ Mean ln(hourly wage) by firm amenity provision (week 90)

Identified Set for Type Probabilities (Week 208)

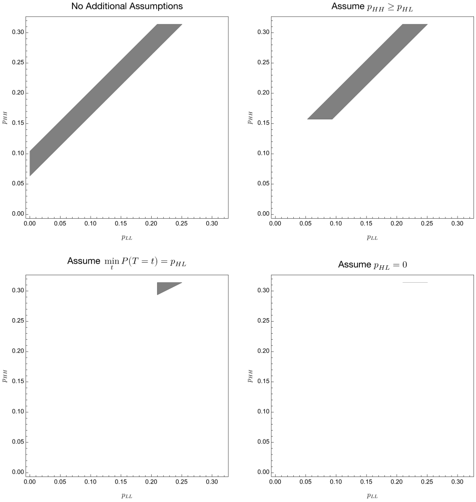


Figure: Identified set for $\Big(\mathbb{P}\big(T = (L, L)\big), \mathbb{P}\big(T = (H, H)\big)\Big)$ (week 208)

◂ Identified set (week 90)

Propensity Scores by Week

	$\mathbb{P}[D = H \mid Z = 0]$	$\mathbb{P}[D = H \mid Z = 1]$	$\mathbb{P}[D = L \mid Z = 0]$	$\mathbb{P}[D = L \mid Z = 1]$
Week 90	0.22	0.24	0.24	0.22
Week 135	0.28	0.30	0.24	0.24
Week 180	0.29	0.33	0.25	0.25
Week 208	0.31	0.36	0.25	0.25

Table: Propensity scores by week (amenity=health insurance)

◂ Identified set (week 90)

Job Corps (Week 208)

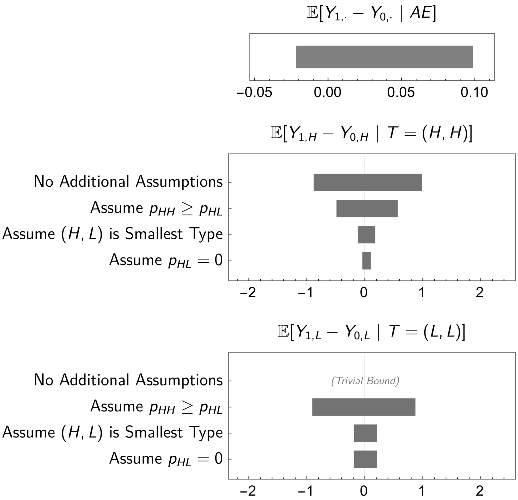


Figure: Bounds in Week 208

◂ Bounds (week 90)

Replication of Lee (2009) Bounds

	$\mathbb{P}[D > 0 \mid Z = 0]$	$\mathbb{P}[D > 0 \mid Z = 1]$	$p \equiv \mathbb{P}(AE)$	$\mathbb{E}[Y_{1,D_1} - Y_{0,D_0} \mid D_0 > 0, D_1 > 0]$ lower	upper
Week 90	0.460	0.460	0.000	0.047	0.048
Week 135	0.517	0.545	0.051	-0.007	0.084
Week 180	0.540	0.583	0.072	-0.033	0.090
Week 208	0.566	0.607	0.068	-0.022	0.099

Table: Lee's bounds[7]

◂ Bounds (week 90)

[7] Slight numerical difference in treatment bounds compared to Lee (2009) arises as Lee uses wage vingtiles to compute bounds and we do not.